

NOS Practice Problem Statements

Spec-driven development practice — Challenge

How this works: pick a problem statement, write a short spec before writing any code, build a working prototype against that spec, then walk through a quick live demo.

Challenge

Lightweight, scope-limited MVPs inspired by well-known B2B SaaS products. These are not full replicas — each spec must explicitly state what's excluded. Focus: judgment on what NOT to build, alongside solid core functionality.

1. Lightweight Helpdesk

Inspired by: Zendesk

Required features:

- Ticket intake form
- Status pipeline: open / pending / resolved
- Assign ticket to an agent
- Flag SLA breach after 24 hours open

Edge case to handle:

What happens to the SLA flag if a ticket moves to 'pending' — does the clock pause or keep running?

Explicitly out of scope:

No multi-tenant support, no email-to-ticket conversion, no live chat.

2. Simple CRM

Inspired by: HubSpot (core CRM)

Required features:

- Contact and company records
- Deal stages: lead → qualified → won/lost
- Activity log per contact

Edge case to handle:

What happens to open deals attached to a contact that gets marked inactive?

Explicitly out of scope:

No email sync, no marketing automation, no reporting dashboards.

3. Minimal Project Board

Inspired by: Jira / Asana

Required features:

- Projects containing tasks
- Status columns (kanban-style)
- Group tasks by sprint/cycle
- Show count of open vs. closed tasks

Edge case to handle:

What happens to a task's sprint assignment when a sprint ends but the task isn't done?

Explicitly out of scope:

No permissions/roles, no time tracking, no integrations.

4. Basic Invoicing Tool

Inspired by: Stripe Invoicing / QuickBooks

Required features:

- Create invoice with line items
- Mark invoice paid / unpaid / overdue
- Show monthly revenue summary

Edge case to handle:

What happens to an invoice marked 'paid' if a line item is edited afterward?

Explicitly out of scope:

No payment gateway integration, no recurring invoices, no tax calculation.

5. Scheduling / Booking Tool

Inspired by: Calendly

Required features:

- Set availability windows
- External party books an open slot
- Prevent overlapping bookings
- Show upcoming bookings list

Edge case to handle:

What happens if the host changes their availability after a slot in that window has already been booked?

Explicitly out of scope:

No payment collection, no calendar sync (Google/Outlook), no team scheduling.

6. Simple Applicant Tracking System

Inspired by: Lever / Greenhouse

Required features:

- Job requisitions
- Candidates per requisition with stage (applied → interview → offer)
- Show stage-conversion counts

Edge case to handle:

What happens to a candidate's stage if the requisition they applied to is closed?

Explicitly out of scope:

No resume parsing, no interview scheduling integration, no offer-letter generation.

7. Internal Knowledge Base

Inspired by: Notion / Confluence

Required features:

- Articles with categories
- Edit history / versioning per article
- Search by title or tag

Edge case to handle:

What happens when two people edit the same article at the same time?

Explicitly out of scope:

No granular permissions, no external sharing, no rich media embeds.

8. Basic Incident Tracker

Inspired by: PagerDuty / Opsgenie

Required features:

- Log incidents with severity
- Assign an owner
- Status: open / mitigated / resolved
- Show time-to-resolution metric

Edge case to handle:

What happens to time-to-resolution if an incident is reopened after being marked resolved?

Explicitly out of scope:

No on-call rotation/paging, no integrations with monitoring tools, no auto-escalation.